42000 Cryptography Implementation Project Ethereum

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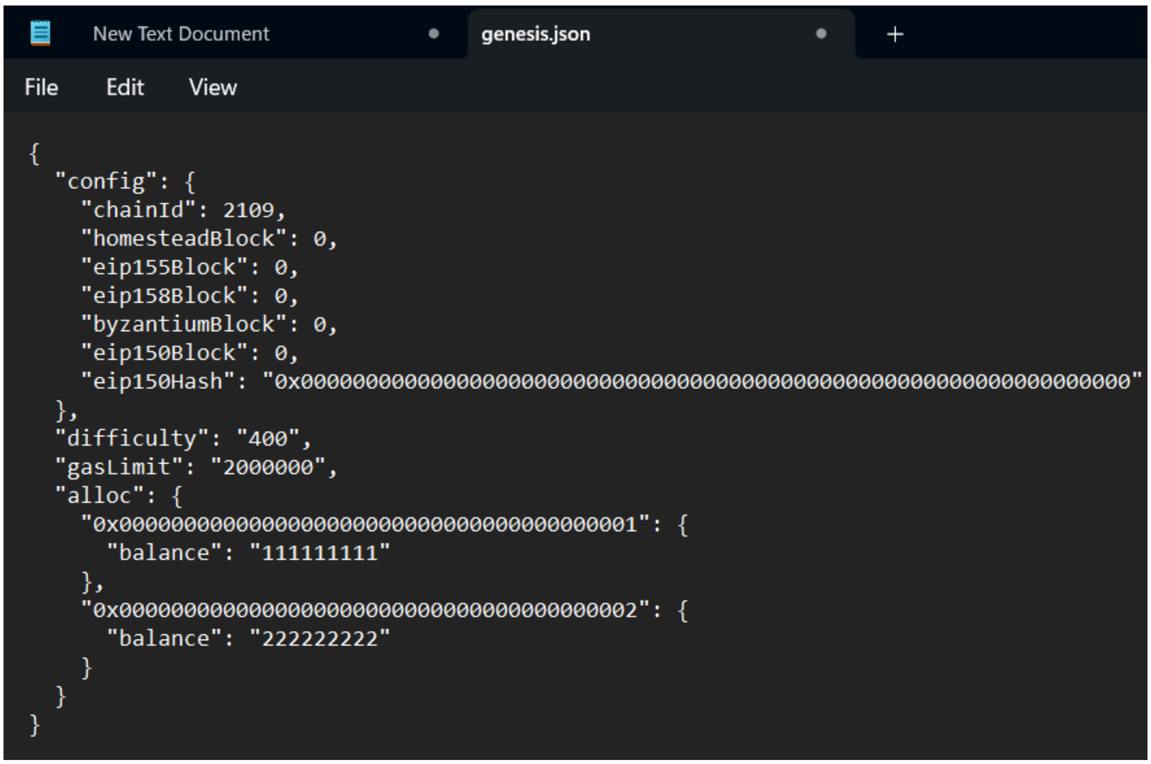
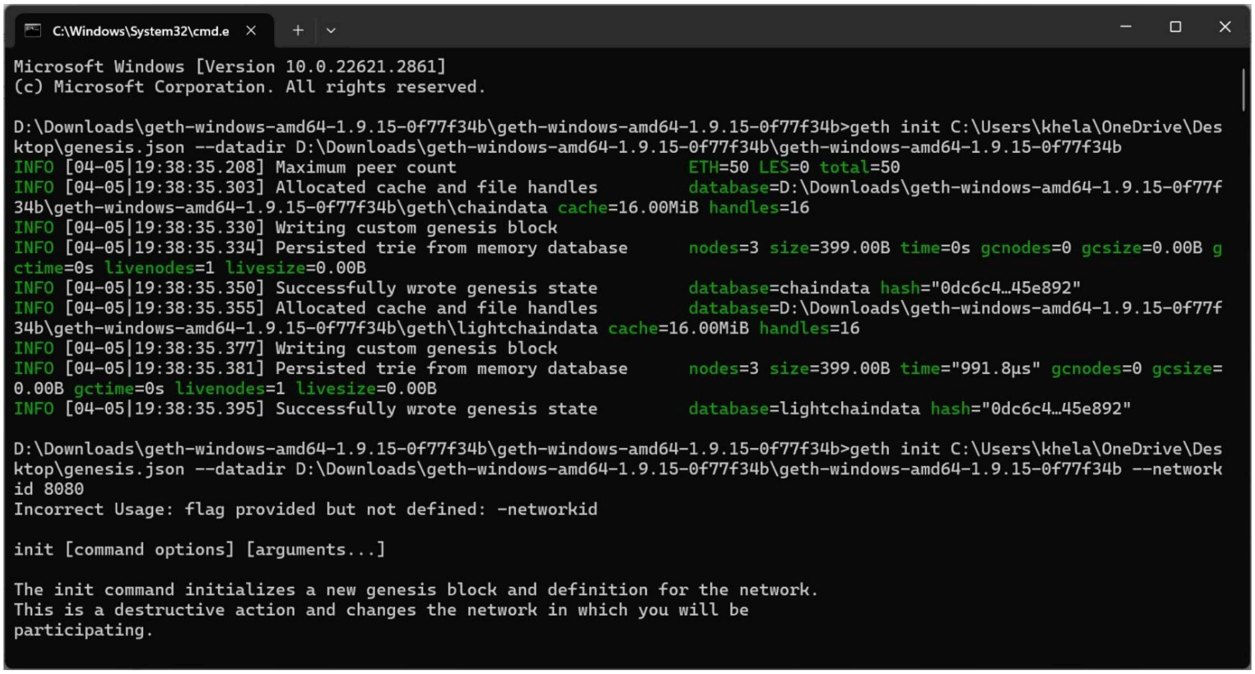
Task: Run your private Ethereum network with a single node.



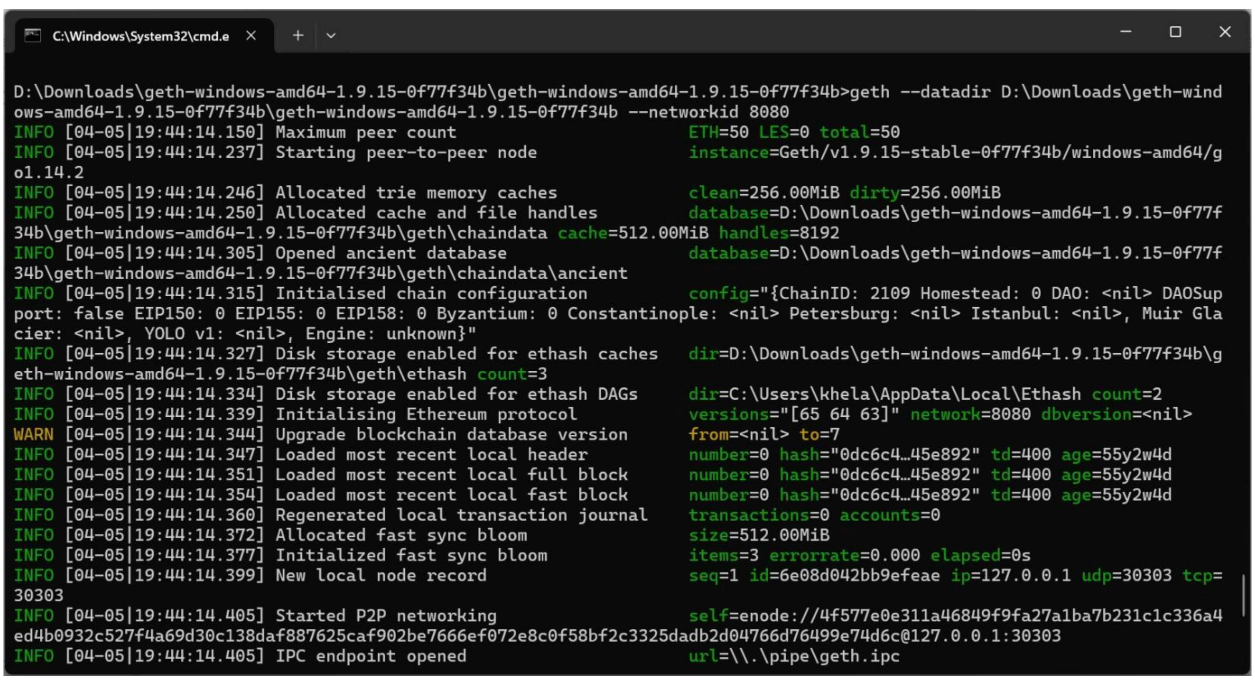

```json
{
  "config": {
    "chainId": 2109,
    "homesteadBlock": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "eip150Block": 0,
    "eip150Hash": "0x0000000000000000000000000000000000000000000000000000000000000000"
  },
  "difficulty": "400",
  "gasLimit": "2000000",
  "alloc": {
    "0x0000000000000000000000000000000000000001": {
      "balance": "111111111"
    },
    "0x0000000000000000000000000000000000000002": {
      "balance": "222222222"
    }
  }
}
```

Fig 1.1 Genesis.json file that provides the software with its initial values.



```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.22621.2861]
(c) Microsoft Corporation. All rights reserved.

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b>geth init C:\Users\khela\OneDrive\Des
ktop\genesis.json --datadir D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b
INFO [04-05|19:38:35.208] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-05|19:38:35.303] Allocated cache and file handles         database=D:\Downloads\geth-windows-amd64-1.9.15-0f77f
34b\geth-windows-amd64-1.9.15-0f77f34b\geth\chaindata cache=16.00MiB handles=16
INFO [04-05|19:38:35.330] Writing custom genesis block
INFO [04-05|19:38:35.334] Persisted trie from memory database      nodes=3 size=399.00B time=0s gcnodes=0 gcsize=0.00B g
ctime=0s livenodes=1 livesize=0.00B
INFO [04-05|19:38:35.350] Successfully wrote genesis state         database=chaindata hash="0dc6c4…45e892"
INFO [04-05|19:38:35.355] Allocated cache and file handles         database=D:\Downloads\geth-windows-amd64-1.9.15-0f77f
34b\geth-windows-amd64-1.9.15-0f77f34b\geth\lightchaindata cache=16.00MiB handles=16
INFO [04-05|19:38:35.377] Writing custom genesis block
INFO [04-05|19:38:35.381] Persisted trie from memory database      nodes=3 size=399.00B time="991.8µs" gcnodes=0 gcsize=
0.00B gctime=0s livenodes=1 livesize=0.00B
INFO [04-05|19:38:35.395] Successfully wrote genesis state         database=lightchaindata hash="0dc6c4…45e892"

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b>geth init C:\Users\khela\OneDrive\Des
ktop\genesis.json --datadir D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b --network
id 8080
Incorrect Usage: flag provided but not defined: -networkid

init [command options] [arguments...]

The init command initializes a new genesis block and definition for the network.
This is a destructive action and changes the network in which you will be
participating.
```

Fig1.2 Initialising the Ethereum network



```
C:\Windows\System32\cmd.e

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b>geth --datadir D:\Downloads\geth-wind
ows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b --networkid 8080
INFO [04-05|19:44:14.150] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-05|19:44:14.237] Starting peer-to-peer node               instance=Geth/v1.9.15-stable-0f77f34b/windows-amd64/g
o1.14.2
INFO [04-05|19:44:14.246] Allocated trie memory caches             clean=256.00MiB dirty=256.00MiB
INFO [04-05|19:44:14.250] Allocated cache and file handles         database=D:\Downloads\geth-windows-amd64-1.9.15-0f77f
34b\geth-windows-amd64-1.9.15-0f77f34b\geth\chaindata cache=512.00MiB handles=8192
INFO [04-05|19:44:14.305] Opened ancient database                  database=D:\Downloads\geth-windows-amd64-1.9.15-0f77f
34b\geth-windows-amd64-1.9.15-0f77f34b\geth\chaindata\ancient
INFO [04-05|19:44:14.315] Initialised chain configuration          config="{ChainID: 2109 Homestead: 0 DAO: <nil> DAOSup
port: false EIP150: 0 EIP155: 0 EIP158: 0 Byzantium: 0 Constantinople: <nil> Petersburg: <nil> Istanbul: <nil>, Muir Gla
cier: <nil>, YOLO v1: <nil>, Engine: unknown}"
INFO [04-05|19:44:14.327] Disk storage enabled for ethash caches   dir=D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\g
eth-windows-amd64-1.9.15-0f77f34b\geth\ethash count=3
INFO [04-05|19:44:14.334] Disk storage enabled for ethash DAGs     dir=C:\Users\khela\AppData\Local\Ethash count=2
INFO [04-05|19:44:14.339] Initialising Ethereum protocol           versions="[65 64 63]" network=8080 dbversion=<nil>
WARN [04-05|19:44:14.344] Upgrade blockchain database version      from=<nil> to=7
INFO [04-05|19:44:14.347] Loaded most recent local header          number=0 hash="0dc6c4…45e892" td=400 age=55y2w4d
INFO [04-05|19:44:14.351] Loaded most recent local full block      number=0 hash="0dc6c4…45e892" td=400 age=55y2w4d
INFO [04-05|19:44:14.354] Loaded most recent local fast block      number=0 hash="0dc6c4…45e892" td=400 age=55y2w4d
INFO [04-05|19:44:14.360] Regenerated local transaction journal    transactions=0 accounts=0
INFO [04-05|19:44:14.372] Allocated fast sync bloom                size=512.00MiB
INFO [04-05|19:44:14.377] Initialized fast sync bloom              items=3 errorrate=0.000 elapsed=0s
INFO [04-05|19:44:14.399] New local node record                    seq=1 id=6e08d042bb9efeae ip=127.0.0.1 udp=30303 tcp=
30303
INFO [04-05|19:44:14.405] Started P2P networking                   self=enode://4f577e0e311a46849f9fa27a1ba7b231c1c336a4
ed4b0932c527f4a69d30c138daf887625caf902be7666ef072e8c0f58bf2c3325dadb2d04766d76499e74d6c@127.0.0.1:30303
INFO [04-05|19:44:14.405] IPC endpoint opened                      url=\\.\pipe\geth.ipc
```

Fig 1.3 Running the network with the console enabled.

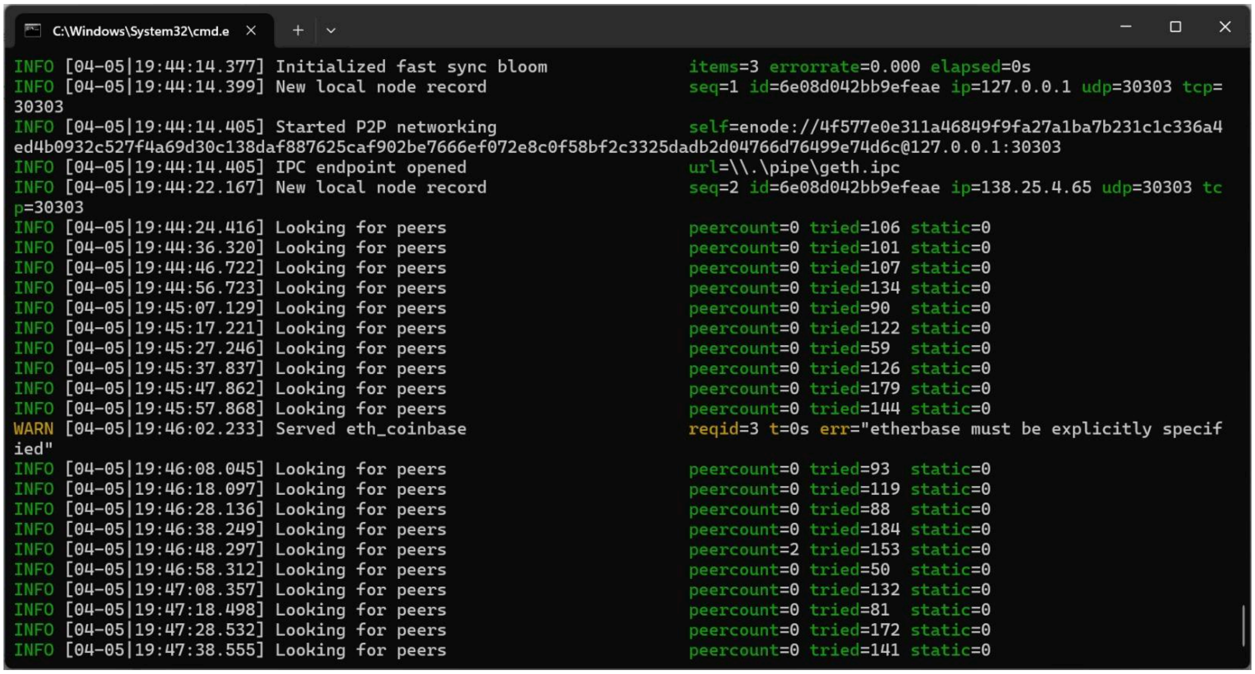


Fig 1.4 Playing with some supported commands.

Task 2: Create accounts and Mining for tokens.

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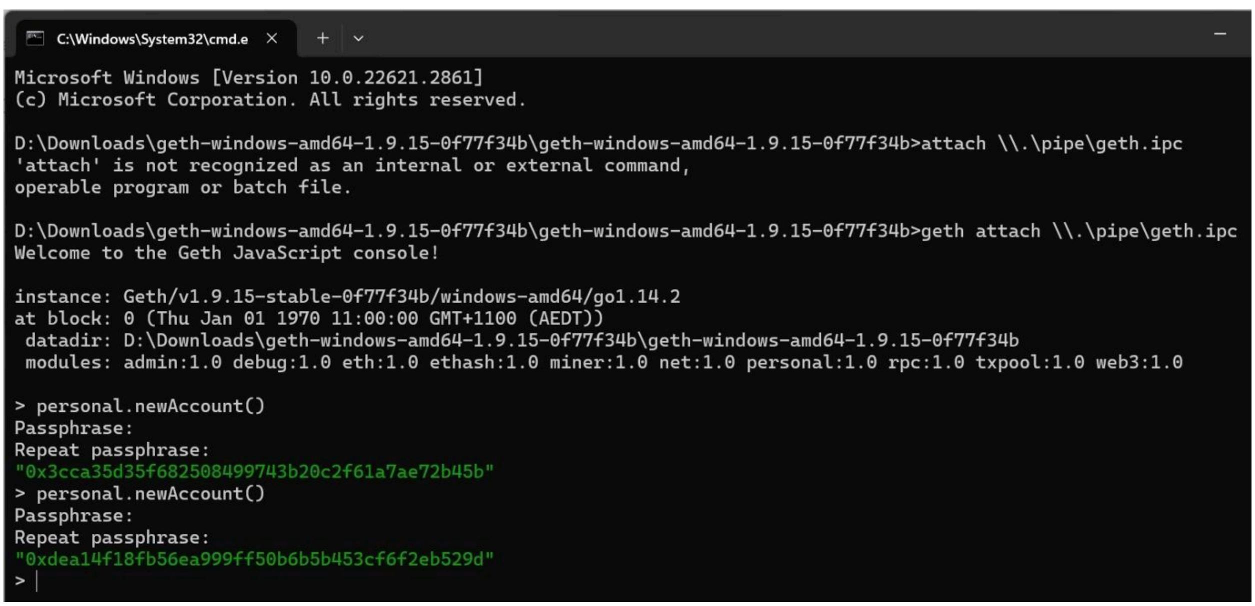


Fig 2.1 Creating two accounts.

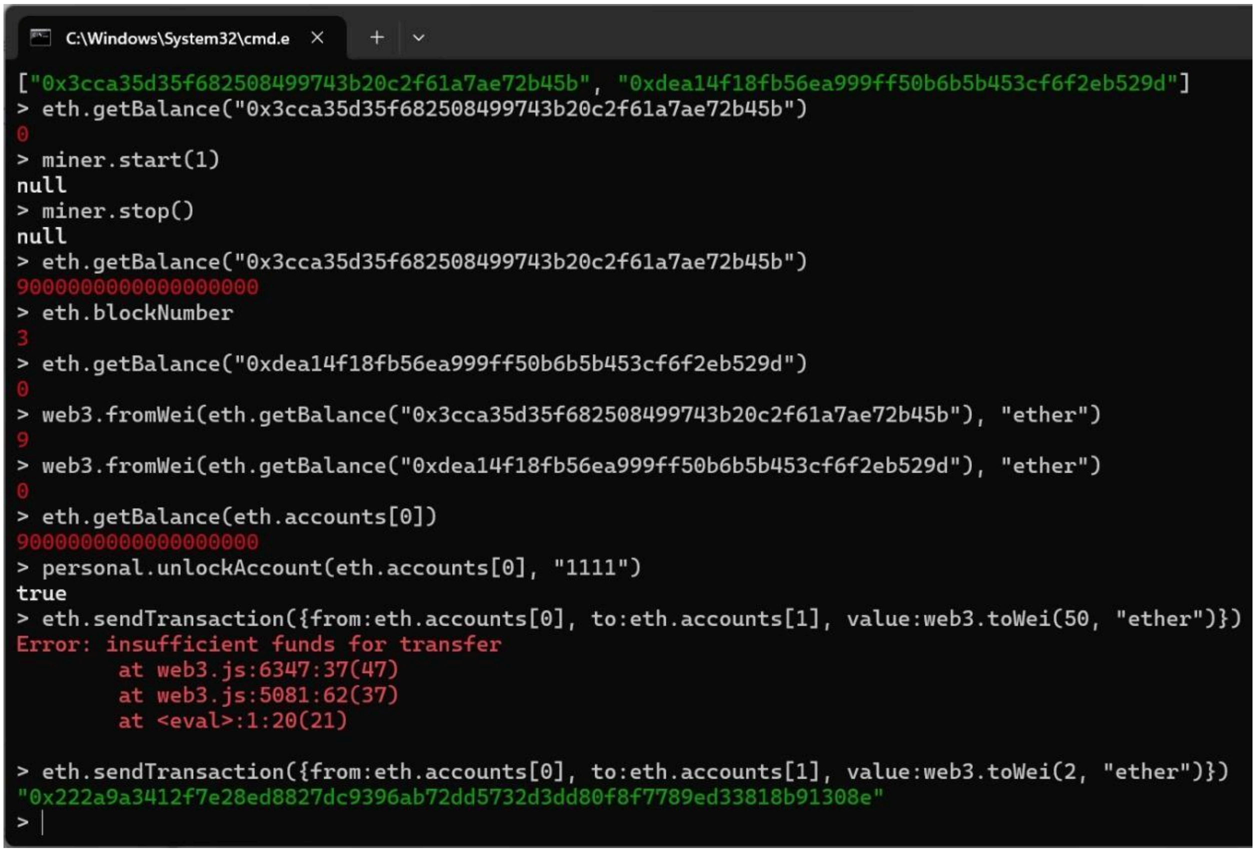
```
C:\Windows\System32\cmd.e

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b>attach \\.\pipe\geth.ipc
'attach' is not recognized as an internal or external command,
operable program or batch file.

D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b>geth attach \\.\pipe\geth.ipc
Welcome to the Geth JavaScript console!

instance: Geth/v1.9.15-stable-0f77f34b/windows-amd64/go1.14.2
at block: 0 (Thu Jan 01 1970 11:00:00 GMT+1100 (AEDT))
 datadir: D:\Downloads\geth-windows-amd64-1.9.15-0f77f34b\geth-windows-amd64-1.9.15-0f77f34b
 modules: admin:1.0 debug:1.0 eth:1.0 ethash:1.0 miner:1.0 net:1.0 personal:1.0 rpc:1.0 txpool:1.0 web3:1.0

> personal.newAccount()
Passphrase:
Repeat passphrase:
"0x3cca35d35f682508499743b20c2f61a7ae72b45b"
> personal.newAccount()
Passphrase:
Repeat passphrase:
"0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
0
> eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d")
0
>
> miner.setEtherbase("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
true
> eth.coinbase
"0x3cca35d35f682508499743b20c2f61a7ae72b45b"
>
```

Fig 2.2 Checking the initial values of both the accounts.

Task 3: Create transaction and mine into blocks

```
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
0
> eth.accounts
["0x3cca35d35f682508499743b20c2f61a7ae72b45b", "0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"]
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
0
> miner.start(1)
null
> miner.stop()
null
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
9000000000000000000
> eth.blockNumber
3
> eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d")
0
> web3.fromWei(eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b"), "ether")
9
> web3.fromWei(eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"), "ether")
0
>
```

Fig 3.1 Balance of both the accounts after mining.



```
C:\Windows\System32\cmd.e
["0x3cca35d35f682508499743b20c2f61a7ae72b45b", "0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"]
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
0
> miner.start(1)
null
> miner.stop()
null
> eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b")
9000000000000000000
> eth.blockNumber
3
> eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d")
0
> web3.fromWei(eth.getBalance("0x3cca35d35f682508499743b20c2f61a7ae72b45b"), "ether")
9
> web3.fromWei(eth.getBalance("0xdea14f18fb56ea999ff50b6b5b453cf6f2eb529d"), "ether")
0
> eth.getBalance(eth.accounts[0])
9000000000000000000
> personal.unlockAccount(eth.accounts[0], "1111")
true
> eth.sendTransaction({from:eth.accounts[0], to:eth.accounts[1], value:web3.toWei(50, "ether")})
Error: insufficient funds for transfer
        at web3.js:6347:37(47)
        at web3.js:5081:62(37)
        at <eval>:1:20(21)

> eth.sendTransaction({from:eth.accounts[0], to:eth.accounts[1], value:web3.toWei(2, "ether")})
"0x222a9a3412f7e28ed8827dc9396ab72dd5732d3dd80f8f7789ed33818b91308e"
>
```

Fig 3.2 Transaction from base account to the second account

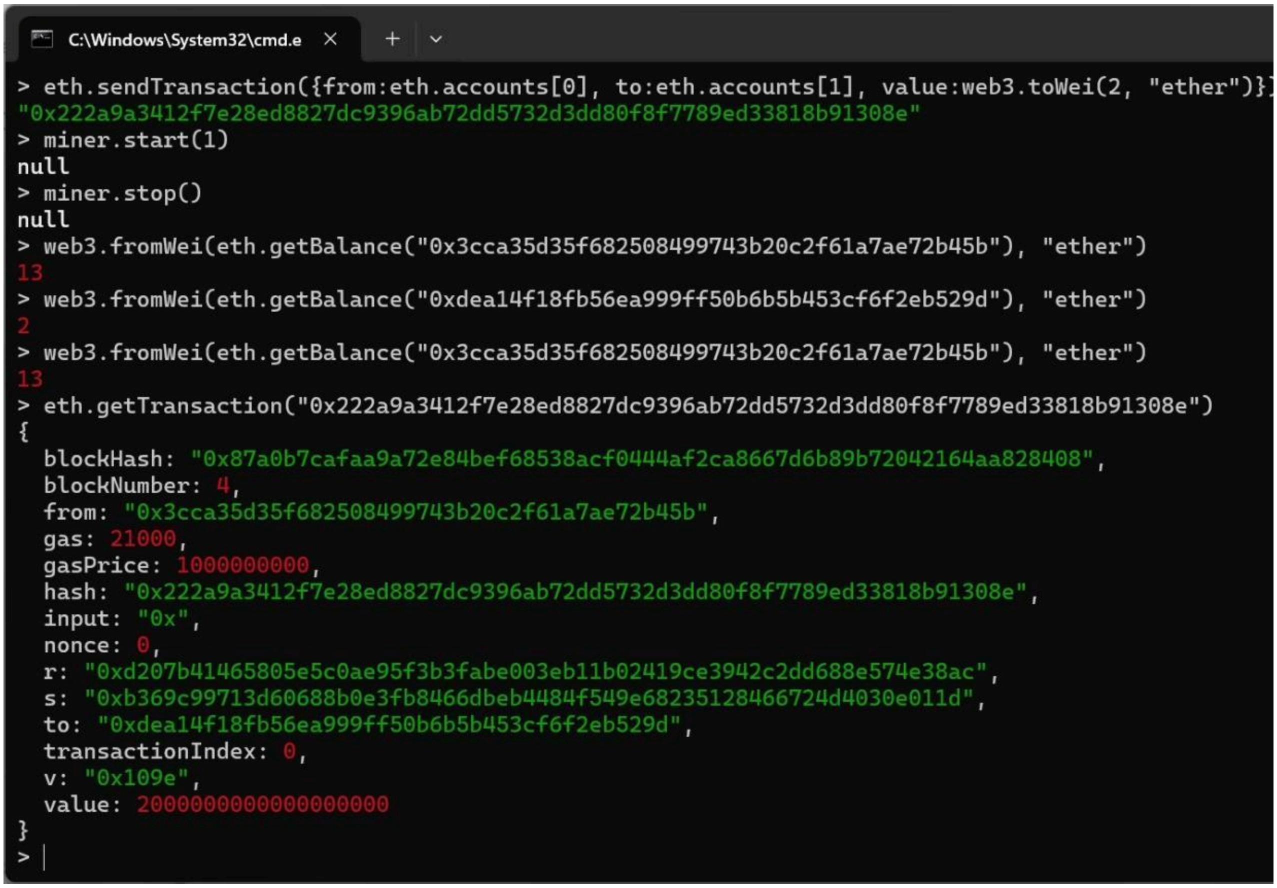


Fig 3.3 Balance of the two accounts and retrieval of transaction with its hash value.

To get things going, we made two accounts: one called the "etherbase" account to enable transactions. We used "miner.setEthbase" to start mining, and then we used "eth.sendTransaction" to start a transaction between the two accounts. The analysis of the transaction is as follows

- The hash value of block number 4, where the transaction occurred, is known as the blockhash.
- From indicates the ethbase account
- The quantity of 21,000 wei was the transaction gas, which denotes the expense of executing a function or transaction on the Ethereum network.
- The "gasPrice" is set at 1,000,000,000 wei, which is equivalent to 1 gwei.
- Hash value indicates the hash of the transcation.
- The input is denoted by "0x" since it refers to the data that was transmitted along with the transaction, which in our case was none.
- Since this is the first transaction, there have never been any others, which explains why the nonce is 0.

- The r and s values are typical outputs of a standard Elliptic Curve Digital Signature Algorithm (ECDSA) signature.
- The recipient account, in this case our second account, is specified via the "To" field.
- Since this is the first transaction, its transaction index is 0
- V is needed to recover the public key.
- The value denotes the amount in wei that was transferred.